



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 2

CANDIDATE
NAME

ANSWER SCHEME

CIVICS
GROUP

1 7 S

INDEX
NUMBER

H2 BIOLOGY

9744/02

Paper 2 Structured Questions

12 September 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use

1	/ 9
2	/ 9
3	/ 11
4	/ 10
5	/ 11
6	/ 9
7	/ 10
8	/ 8
9	/ 12
10	/ 11
Total	/ 100

QUESTION 1

Fig. 1.1 is an electron micrograph of part of a eukaryotic cell.



Fig. 1.1

(a) Identify the structures **J** and **K**. [2]

- **J:** Crista
- **K:** Mitochondrial matrix

(b) Describe two structural features shown in Fig. 1.1 that identify **G** as the Golgi apparatus and **not** the rough endoplasmic reticulum. [2]

- **G** has no ribosomes while the rough ER is attached with ribosomes
- **G** has flattened membranous sacs that are stacked / no connection between membranes while the rER membranes are interconnected
- **G** is not continuous with outer membrane of nuclear envelope while the rER is
- **G** has vesicles at ends of sacs / swellings at end of sacs (for vesicle formation) while there are no ends for vesicles formation in the rER
- **G** has a slight curvature compared to the rER

The Golgi apparatus is an organelle that is important for many functions. One example is the formation of lysosomes.

Before lysosomal hydrolases can function in the lysosome, they must be sorted from other proteins. Fig. 1.2 shows how a glycosylated lysosomal hydrolase is sorted at the trans face of the Golgi apparatus.

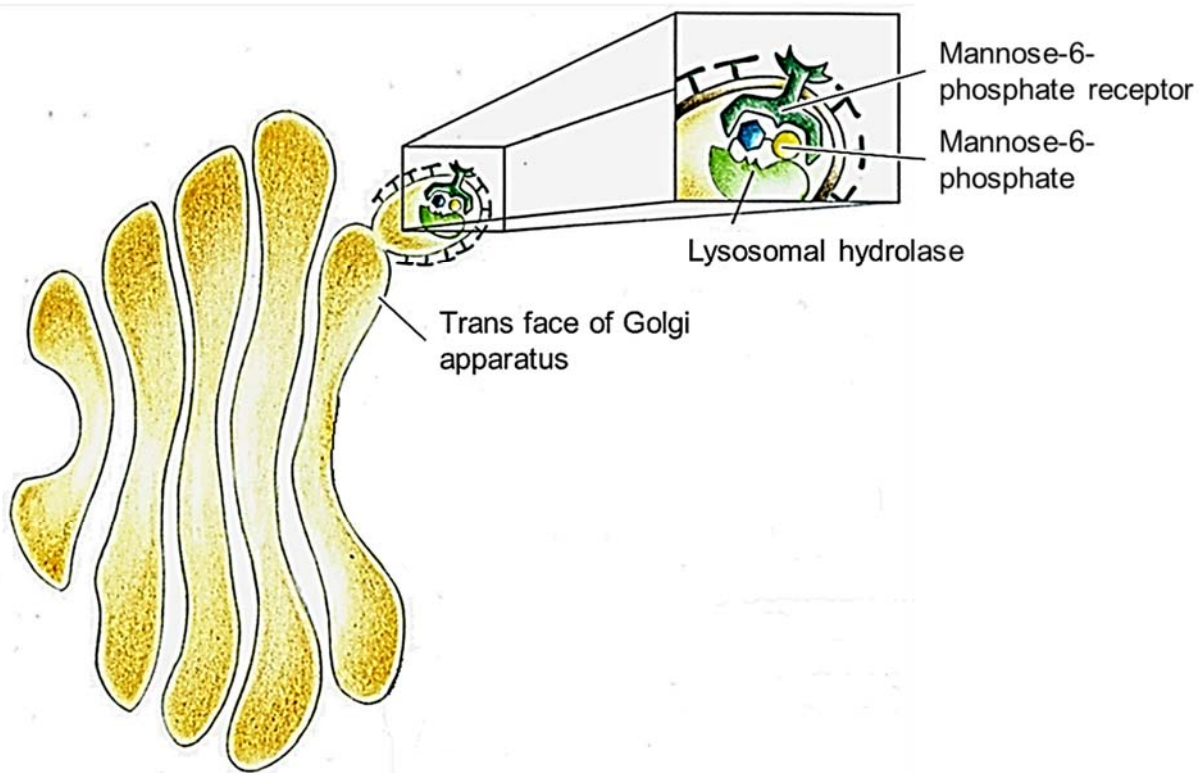


Fig. 1.2

(c) Outline the pathway taken by the lysosomal hydrolase from its site of synthesis to the trans face of the Golgi apparatus. [3]

- After being synthesised by the **ribosome** on the rER, the lysosomal hydrolase enters the **rER lumen**
- **ER vesicle** carrying lysosomal hydrolase **buds off** from rER and travels towards Golgi apparatus
- ER vesicle **fuses** with **cis face** of Golgi apparatus
- Repeated **budding and fusing of vesicles** transports hydrolase to the trans face

(d) With reference to Fig. 1.2, suggest how the Golgi apparatus is able to specifically sort the lysosomal hydrolase from other proteins. [2]

- **Idea that** Only the lysosomal hydrolase contains mannose-6-phosphate / other proteins are not glycosylated / not glycosylated with mannose-6-phosphate

Mannose-6-phosphate is **complementary** in shape to and **binds** to the **mannose-6-phosphate receptor**, allowing the lysosomal hydrolase to be packaged into the

[Total: 9]

QUESTION 2

(a) Contrast the processes of facilitated diffusion and active transport.

[3]

- Facilitated diffusion involves transport of substances **down a concentration gradient** whereas active transport occurs **against a concentration gradient**.
- Facilitated diffusion **does not require ATP** whereas active transport **requires ATP**.
- Facilitated diffusion involves **channel or carrier proteins** whereas active transport only involves **protein pump**.

(b) A group of students investigated the uptake of chloride ions in barley plants. They divided the plants into two groups and placed their roots in solutions containing radioactive chloride ions.

- Group **A** plants had a substance that inhibited respiration added to the solution.
- Group **B** did not have the substance added to the solution.

The students calculated the total amount of chloride ions absorbed by the plants every 15 minutes. Their results are shown in Fig. 2.1.

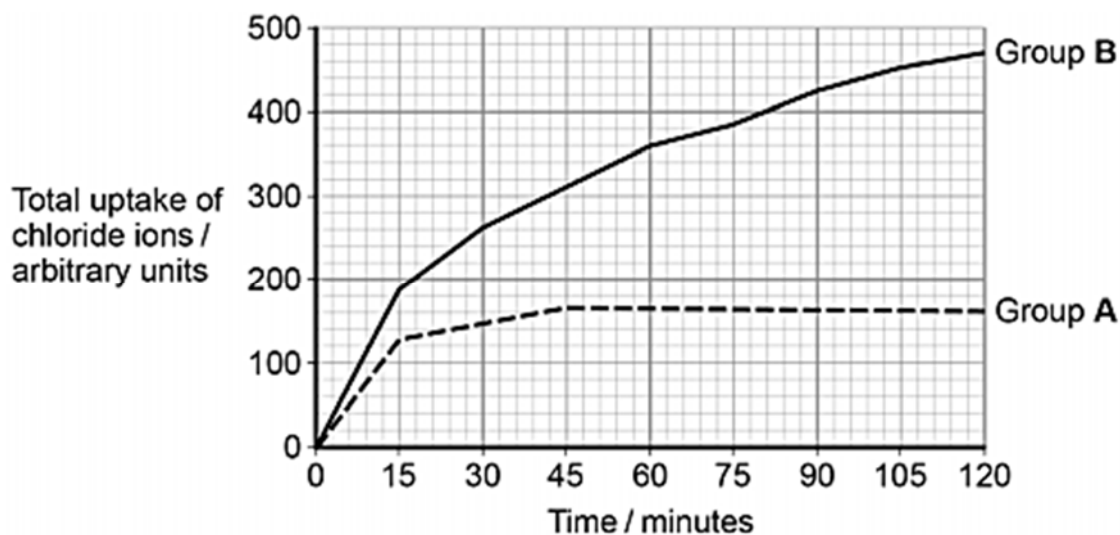


Fig. 2.1

(i) Calculate the ratio of the **rate** of uptake of chloride ions in the first hour to the **rate** of uptake of chloride ions in the second hour for group **B** plants. [2]

- rate of uptake in the first hour = $\frac{360 - 0}{60}$
= 6 au min⁻¹
- rate of uptake in the second hour = $\frac{470 - 360}{60}$
= 1.83 au min⁻¹
- Ratio = 6 / 1.83 = **3.3 : 1**

(ii) Explain the results shown in Fig. 2.1.

[4]

- In Group A, **from 0-15 mins**, the initial rate of uptake of chloride ions is slower as only **facilitated diffusion** occurred.
- In Group A, **from 45-120 mins**, the total uptake of chloride ions **levels off/plateaus** because the concentrations of chloride ions inside cells and outside cells is the **same/reached equilibrium**.
- In Group B, **from 0-15 mins**, the initial rate of uptake of chloride ions is faster as both **facilitated diffusion** and **active transport** occurred.
- In Group B, **from 15-120 mins**, the total uptake of chloride ions continued to increase because the uptake of chloride ions is **against concentration/did not reach an equilibrium**.
- In Group B, **from 15-120 mins**, the rate of uptake **slows down** as **fewer chloride ions in external solution/ respiratory substrate is used up**.

[Total: 9]

QUESTION 3

Bacteria synthesize the amino acid isoleucine from threonine in a series of five enzyme-catalysed steps. When the concentration of isoleucine increases in the cell, isoleucine inhibits threonine deaminase, the first enzyme in the pathway.

Fig. 3.1 shows these five steps in the pathway to synthesise isoleucine.

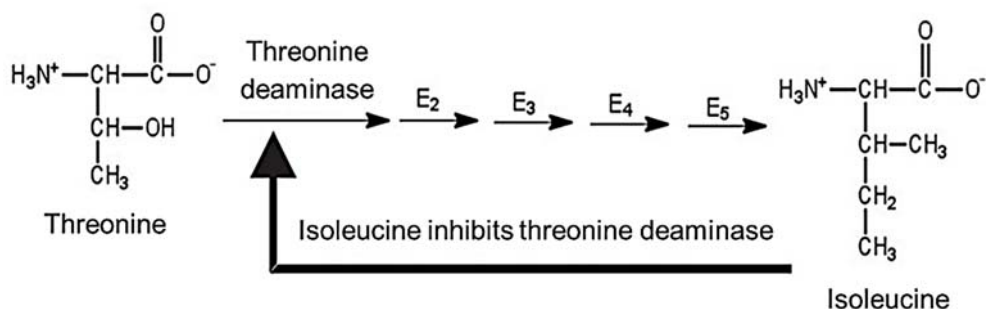


Fig. 3.1

(a) Explain how and why isoleucine inhibits threonine deaminase when the concentration of isoleucine increases. [4]

- Allosteric / feedback / end-product inhibition / isoleucine acts as an allosteric inhibitor of threonine deaminase
- Isoleucine binds to the allosteric site of threonine deaminase, changing its 3D conformation
- Threonine deaminase's active site is no longer complementary in shape to its substrate / the next intermediate in the pathway
- To stop synthesis of isoleucine when there are sufficient amounts in the cell.

These amino acids are used to synthesise the peptidoglycan cell walls of bacteria. Peptidoglycans are long chains of the sugars N-acetylmuramic acid (NAM) and N-acetylglucosamine (NAG) which alternate along the chain. A short peptide chain is attached to each NAM and these form cross-links with similar peptide chains from adjacent strands.

Fig. 3.2 shows a diagram representing part of a peptidoglycan structure.

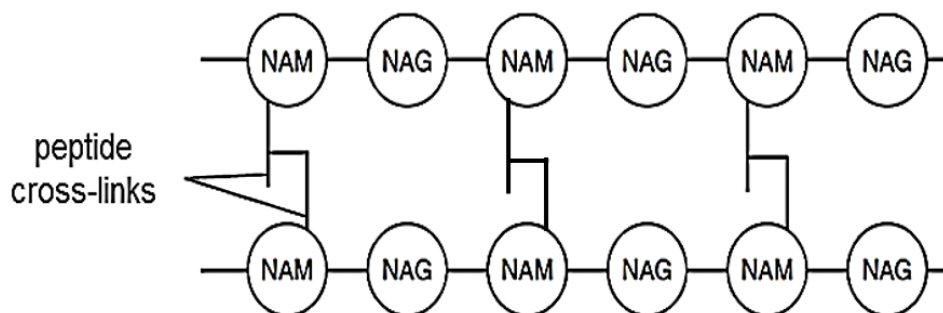


Fig. 3.2

(b) State two differences between the structure of the peptidoglycan cell wall and that of the plant cell wall. [2]

- Monomer: NAM & NAG vs β -glucose
- Cross-links: peptide chains vs. hydrogen bonds

The formation of these peptidoglycan cross-links are catalysed by the enzyme transpeptidase. Penicillin inhibits the action of this enzyme.

(c) Describe the mode of action of penicillin on bacteria. [3]

- Penicillin inhibits the formation of peptidoglycan cross-links in the bacterial cell wall by binding the penicillin β -lactam ring to transpeptidase.
- As the cell continues to grow, the cell wall becomes weaker
- *Idea that* the cell eventually bursts from osmotic pressure and dies

Bacteria may be classified as Gram-positive or Gram-negative bacteria.

Fig. 3.3 shows a diagram of part of the cell walls of both Gram-positive and Gram-negative bacteria.

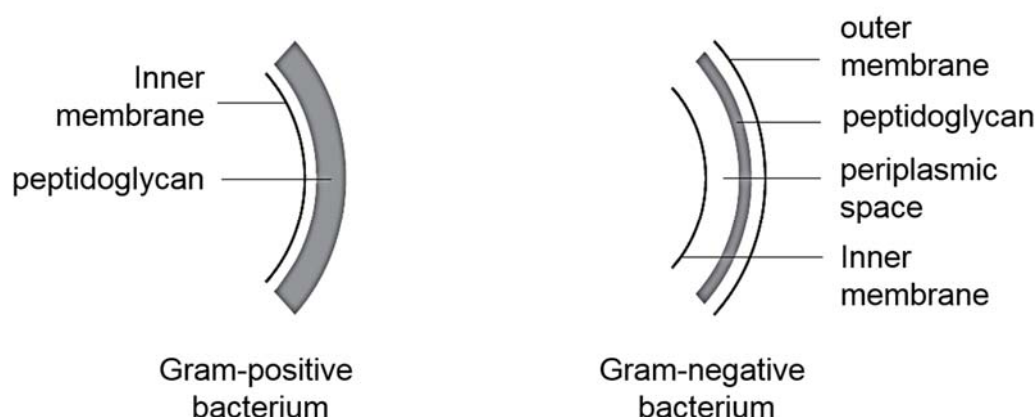


Fig. 3.3

(d) Predict the effectiveness of penicillin on killing the two types of bacteria. Explain your answer. [2]

- Penicillin is more likely to kill Gram-positive bacteria
- Gram-negative bacteria have an outer membrane that may prevent penicillin from accessing the peptidoglycan layer.

OR

- Gram-positive bacteria have a higher proportion of peptidoglycan in its cell wall than Gram-negative bacteria, hence the cell wall would be too weak if there was penicillin

[Total: 11]

QUESTION 4

Palisade cells have both chloroplasts and mitochondria. Exchanges between a mitochondrion, a chloroplast and the cytoplasm surrounding them are shown in Fig 4.1.

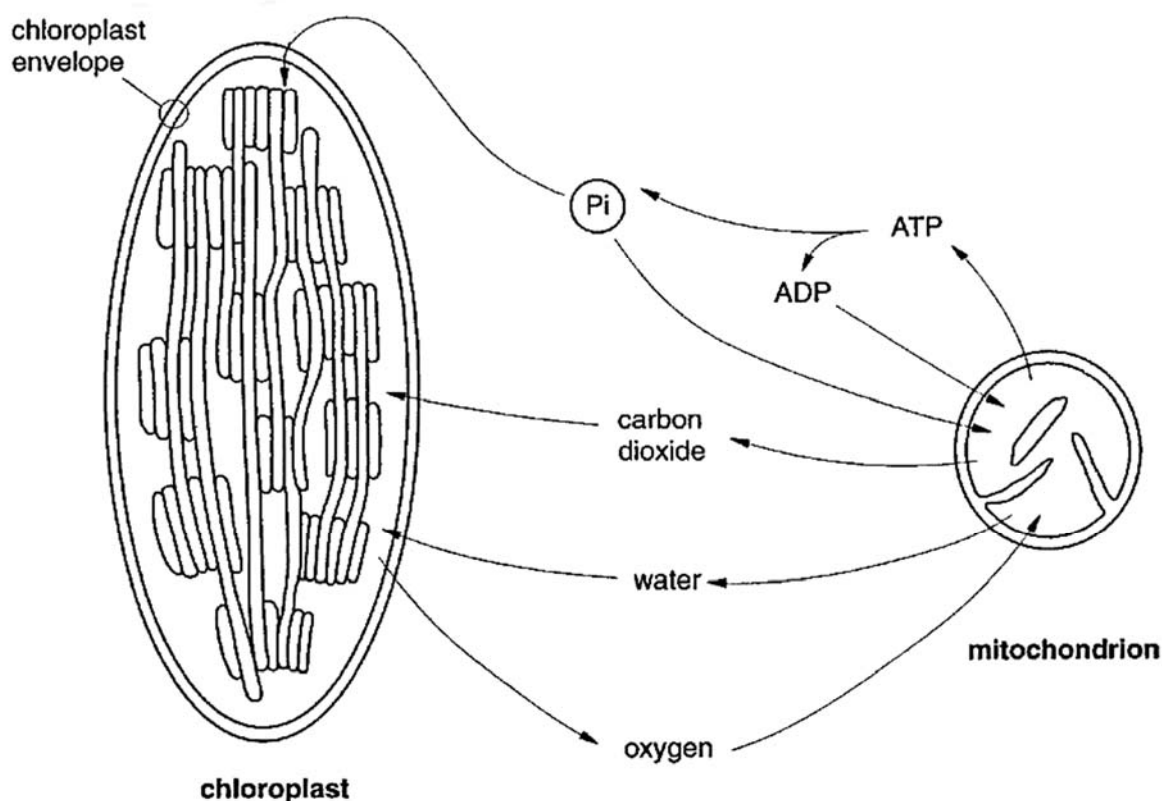


Fig. 4.1

(a) Explain why oxygen is needed for the production of ATP in the mitochondrion. [4]

- Oxygen serves as the final electron acceptor of the electron transport chain and is reduced to form water.
- This allows electrons to be transferred along electron carriers of progressively lower energy levels.
- Thus releasing energy for proton pumps to actively transport protons from the mitochondrial matrix to the intermembrane space
- Forming a proton gradient across the inner mitochondrial membrane.
- Chemiosmosis / facilitated diffusion of protons through ATP synthase can then occur, generating energy for the synthesis of ATP from ADP and Pi.

A leafy shoot is sealed inside a transparent container. The concentration of oxygen in the atmosphere within this container is measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

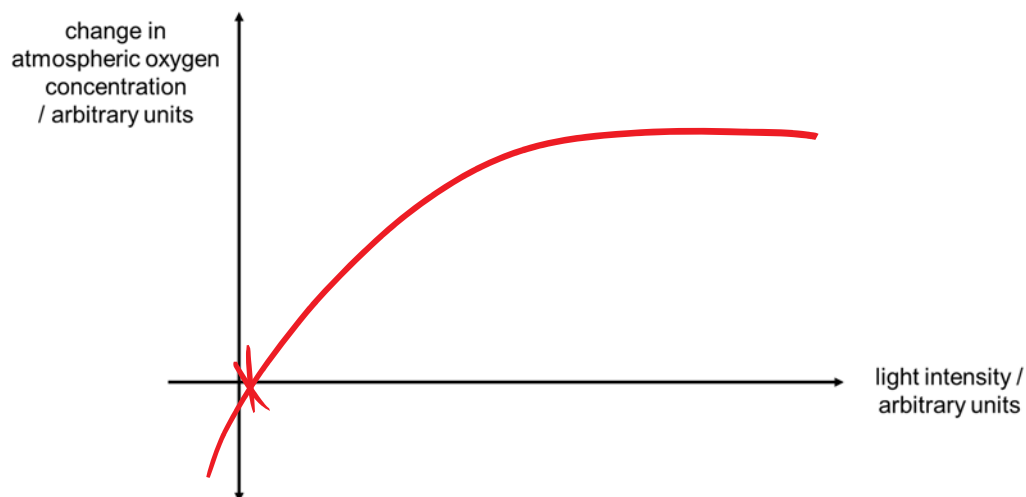
(b) Using Fig. 4.1, explain how it is possible for the oxygen concentration to remain constant. [2]

- At compensation point [1], the rate of photosynthesis and respiration are the same[1].
- OR
- At compensation point [1], the rate of oxygen taken in by respiration is the same as the rate given out by photosynthesis / no net change in the atmospheric concentration of oxygen[1].

(c) (i) Sketch using the axes below, a graph showing the relationship between light intensity and the change in atmospheric oxygen concentration in the container with the leafy shoot. [1]

(ii) Indicate with an 'X' the light intensity where the oxygen concentration remains constant. [1]

[Accept: label line for X]



(d) Both the chloroplast and mitochondrion synthesise ATP.

Describe the fates of ATP synthesised by these two organelles. [2]

- (idea that) ATP synthesised by the mitochondrion enter the cytosol to be used for energy-requiring cellular activities...
- ...while ATP synthesised by the chloroplast enter the Calvin cycle in the stroma and is used to phosphorylate glycerate-3-phosphate to form glycerate-1,3-bisphosphate / regenerate RuBP

[Total: 10]

QUESTION 5

Fig. 5.1 shows the stages involved in the synthesis of a p53 protein in a eukaryotic cell.

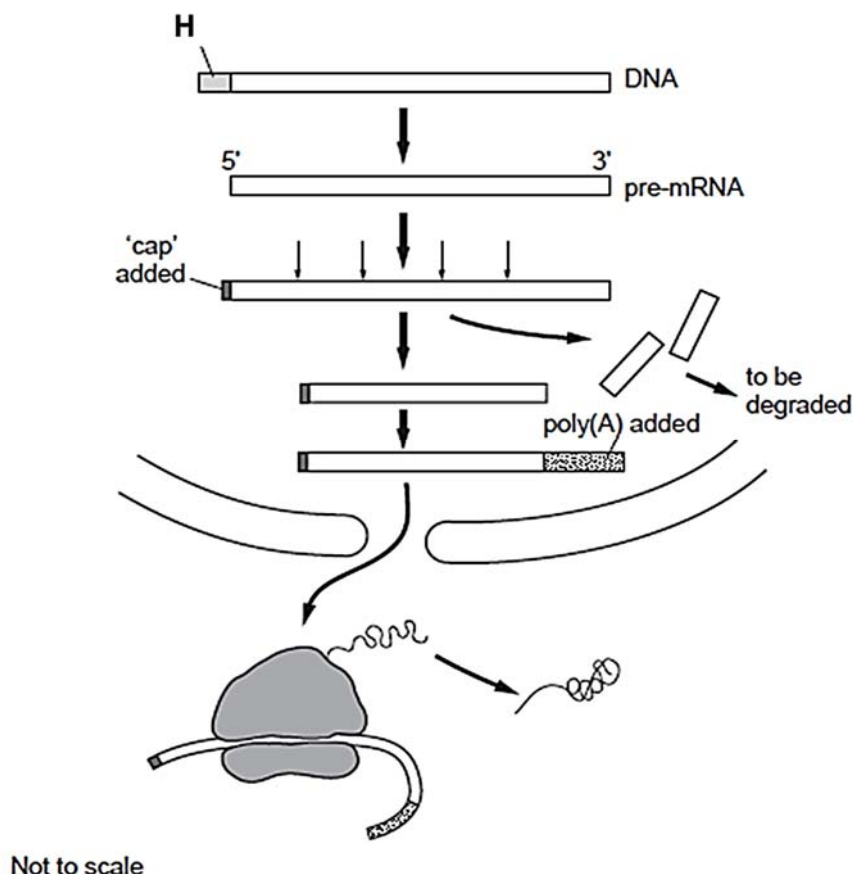


Fig. 5.1

(a) Name the structure H.

[1]

- Promoter (R: deoxyribonucleic acid)

(b) Outline the events that occur in the nucleus of the cell to produce a functional molecule of mRNA encoding a p53 protein.

[4]

- RNA polymerase reads the template strand from 3' to 5' to form pre-mRNA / adds complementary ribonucleotides to the 3'OH end of the pre-mRNA strand
- 7-methylguanose / modified guanine cap added to the 5' end of the pre-mRNA strand
- Introns are excised and exons are spliced
- 3' poly(A) tail added

(c) In a particular defective cell, all the pre-mRNA encoding the p53 protein were **not** cut.

(i) Explain how this would affect the function of the p53 protein. [4]

- Introns which are non-coding would be translated (along with the exons)
- If the number of nucleotides in introns are not in multiples of three, there would be a shift in the reading frame by the ribosome
OR There may be a premature stop codon which produces a truncated protein
OR There may also be a longer protein produced if the stop codon is removed
- These will lead to a change in the sequence of amino acids / primary structure of p53 protein
- And thus different folding to give a different 3D conformation of the p53 protein, which may give rise to a non-functional p53 protein

(ii) Explain why this may not lead to cancer. [2]

- Cancer is a multi-step process
- The cell would need additional gain-of-function mutations to the proto-oncogenes / accumulation of mutations to a single cell line

[Total: 11]

QUESTION 6

STAT proteins are cytoplasmic transcription factors that play important roles in the development and differentiation of many cell types. Upon external stimulation, STAT protein is activated from its inactive form and binds to another activated STAT protein to form a dimer. This protein dimer then translocates to the nucleus and regulates the expression of other genes as shown in Fig. 6.1.

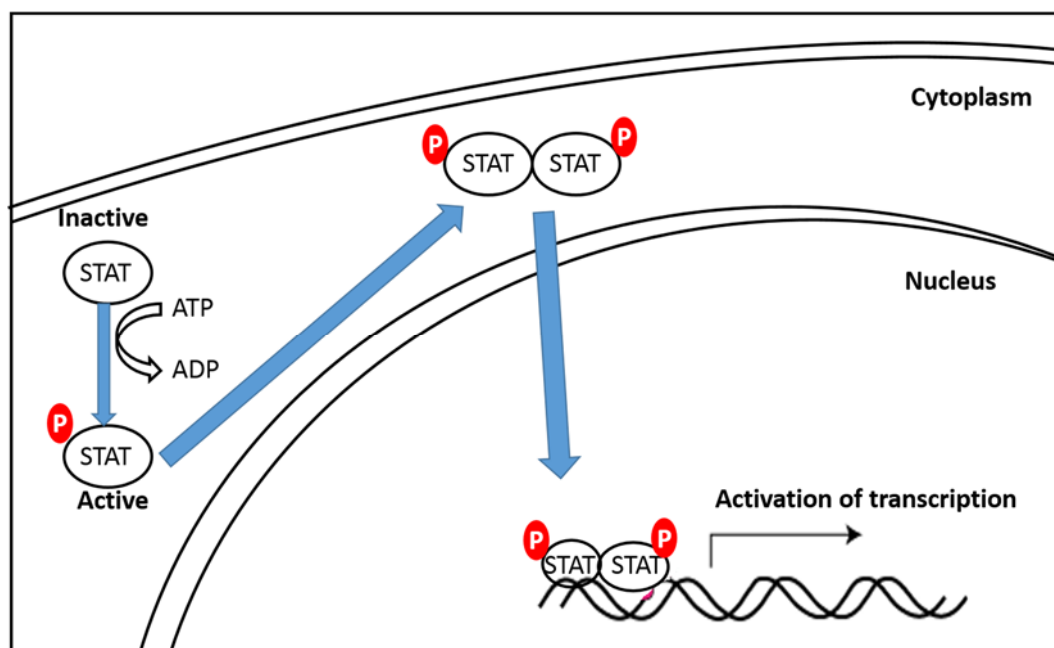


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) explain how the inactive STAT protein is converted to its active form. [2]

- Via Post-Translational modification / phosphorylation
- ATP is hydrolysed to donate a phosphate group to the inactive STAT to form an active STAT protein.

(ii) besides converting the inactive form of STAT protein to its active form, describe how the level of the inactive STAT protein may be controlled after its production. [2]

- These inactive forms of STAT may be tagged with ubiquitin proteins which are recognized by and degraded in proteasomes
- This decreases concentration / amount of STAT

STAT5 gene, a member of the *STAT* family, is widely expressed in hematopoietic stem cells (HSC) to regulate the self-renewal and differentiation of the stem cells.

Fig. 6.2 shows the differentiation of HSC leading to the formation of different cell types such as T cell and B cell.

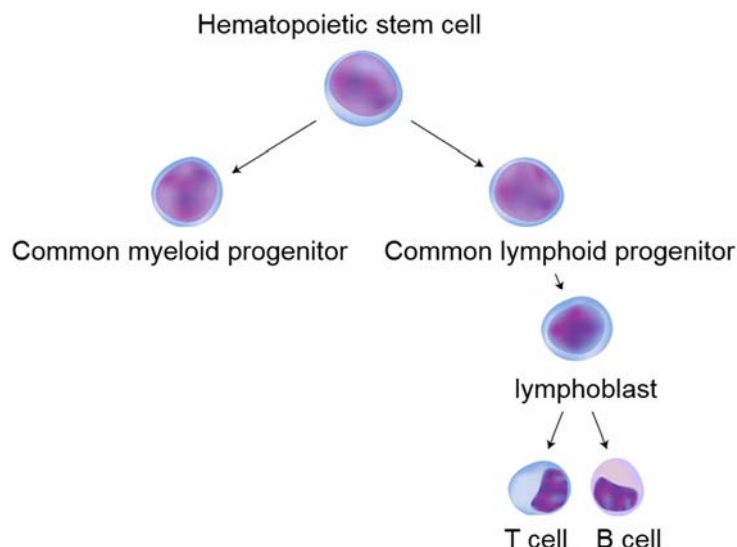


Fig. 6.2

(b) Explain how the different cell types such as T cell and B cell can arise from a single hematopoietic stem cell. [3]

- *Idea of differential gene expression* occurred during differentiation.
- The specific combination of activators present in the cells are different. They bind to their respective enhancers to up-regulate transcription of T cell -specific genes and B cell-specific genes respectively / *idea of repressors and silencers*

Or

- *Idea of* Different sets of cell type specific genes were switched on/off by DNA methylation / histone deacetylation / histone methylation....
- Different sets of proteins are synthesized, causing the two cells to have different structures and hence functions.

(c) In humans there are different forms of *STAT5* protein, each plays a slightly different role in different cell types.

Explain how the same *STAT5* gene can produce different forms of *STAT5* protein. [2]

- Alternative RNA splicing
- Different spliceosomes in different cell types bind to specific splicing sites in the introns of pre-mRNA
- Different combinations of exons are spliced together to form three different mature mRNA hence different protein forms of *STAT5* protein.

[Total: 9]

QUESTION 7

Apart from animal cells, bacteria such as *E. coli* can also be infected by viruses. An example of such viruses is the T4 bacteriophage.

Fig. 7.1 shows a T4 bacteriophage which has been modified to contain the DNA of Mu bacteriophage, a temperate phage. The DNA of Mu bacteriophage is able to integrate into the bacterial chromosome.

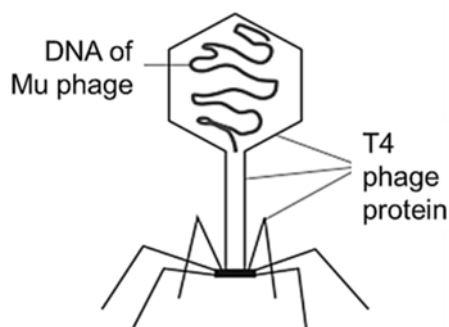


Fig. 7.1

(a) (i) Describe how DNA of Mu bacteriophage enters an *E. coli* host cell. [2]

- Tail fibres recognize and bind to specific / complementary receptors on *E. coli* cell surface membrane
- Contraction of the contractile sheath results in injection of the Mu phage DNA into the host cell

(ii) Explain the importance of the phage genome in integrating the phage DNA into the host chromosome. [2]

- Has genes that code for proteins/enzymes ...
- which break both phage circular DNA and bacterial chromosome and join them to each other

(b) Describe how the reproduction of these modified T4 bacteriophages could contribute to genetic variation in a colony of *E. coli*. [4]

- Specialised transduction
- During the excision of Mu prophage from the *E.coli* genome, imprecise excision causes adjacent *E.coli* genes to be excised together with the Mu prophage.
- This prophage with adjacent bacterial genes are encapsidated to form a specialized transducing phage particle
- When this phage particle infects another *E.coli*, the bacterial DNA from the previous *E.coli* and the phage genome is injected into its new bacterial host.
- The *E.coli* genes can subsequently replace the homologous region of the new host's cell by homologous recombination.

(c) Some scientists believe that bacteriophages could be used to treat people with bacterial infection.

Suggest the properties of the bacteriophages that would make this possible. [2]

- Bacteriophages only infect bacteria / do not infect human cells
- Bacteriophages can lyse the bacterial host cell
- Do not give side effects / allergic responses to humans
- Able to replicate inside bacterial hosts to produce more bacteriophage for treatment

[Total: 10]

QUESTION 8

A researcher created an artificial plasmid that places the promoter (P_{trp}) and operator (O_{trp}) of the *trp* operon upstream of the *lac* structural genes and downstream of the CRP binding site of the *lac* operon, as shown in Fig. 8.1. The *trp R* gene, which codes for the trp repressor, is also inserted into the plasmid.

The plasmids were introduced into bacterial cells.

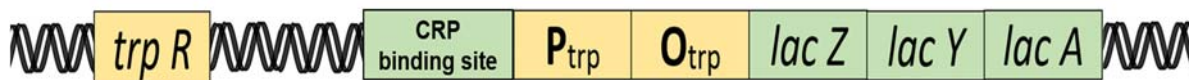


Fig. 8.1

(a) State whether the above operon is an inducible or a repressible system. [1]

- Repressible

(b) Explain the condition needed for the expression of *lac ZYA* genes. [4]

- Absence of the amino acid **tryptophan**
- *Trp R* codes for an inactive repressor that does not bind to O_{trp} as the shape of repressor is not complementary to the operator sequence
- In the absence of tryptophan, which is a co-repressor, repressor remains inactive
- RNA polymerase binds to P_{trp} , transcribing the *lac ZYA* genes

(c) With reference to Fig. 8, describe how this operon is positively regulated. [3]

KU-2

- Idea of up-regulation of *lac* structural genes/ *lac ZYA* gene
- CAP-cAMP complex binds to the CRP binding site
- causes the DNA to bend and enhances the binding of RNA polymerase to the promoter

[Total: 8]

QUESTION 9

(a) Explain the meaning of the terms

[2]

(i) *locus*,

- The **position** occupied by a gene on a chromosome.

(ii) *heterozygous*.

- It is the diploid condition at which the **alleles at a given locus** are **different**.

Immunodeficiency disorders are caused when the immune system's ability to fight infectious diseases is compromised. Bruton's syndrome is a rare immunodeficiency disorder that is characterized by a mutation in the Bruton's tyrosine kinase gene, which **blocks the development of B cell**. Individuals with Bruton's syndrome are vulnerable to repeated, potentially fatal, bacterial infections. To better understand the inheritance pattern of Bruton's syndrome, a family was followed for three generations as shown in Fig. 9.1.

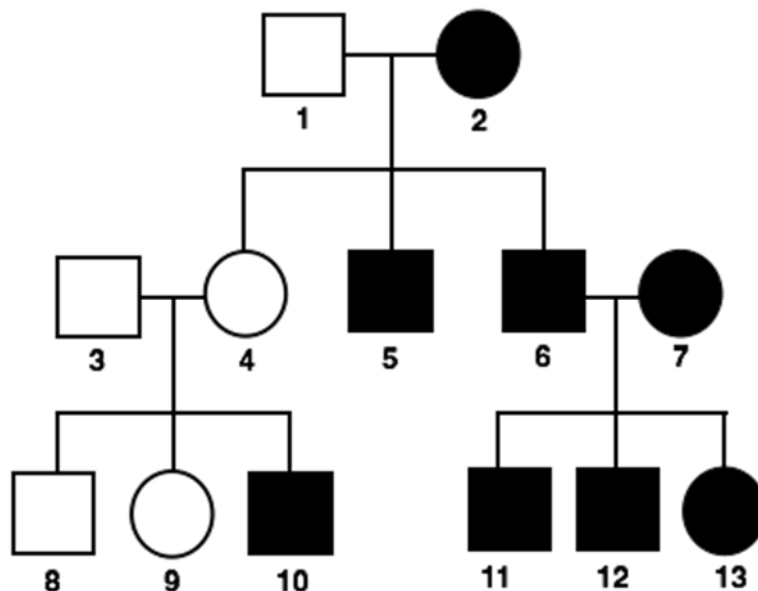


Fig. 9.1

(b) With reference to Fig. 9.1, predict and explain the most likely mode of inheritance of Bruton's syndrome. [3]

- sex-linked recessive

[Explain why recessive]

- *idea of* An affected child can have unaffected parents who passed the recessive alleles to the child. **(cite data)** Individual 10 is affected but not his parents.

[Explain why sex-linked]

- *idea of* All sons of an affected mother are affected **(cite data)** Both sons (individual 5 and 6) of individual 2 are affected

(c) Individual 8 married a normal female whose father is affected.

(i) Using suitable symbols, construct a genetic diagram of the cross. [5]

Key : X^B represents an X chromosome with the dominant normal allele
 X^b represents an X chromosome with the recessive allele for Bruton's syndrome
Y represents the Y chromosome [1]

Parental phenotypes:	normal female whose father is affected	x	normal male	
Parental genotypes:	$X^B X^b$	x	$X^B Y$	[1]
Gametes produced by parents:	X^B X^b	x	X^B Y	[1]
Genotypes of offspring:	$X^B X^B$	$X^B X^b$	$X^B Y$	$X^b Y$ [1]
Phenotypes of offspring:	normal female	normal female	normal male	affected male
Phenotypic ratio:	2 normal female : 1 normal male : 1 affected male			[1]

(ii) State the probability that their first 2 children are all unaffected females. Show your working. [1]

$$(\frac{1}{2} \times \frac{1}{2}) = \frac{1}{4}$$

(d) Children with Bruton's syndrome have no or very low level of immunoglobulins in their blood. As such, they are vulnerable to repeated infections.

Suggest a possible long-term treatment for children with Bruton's syndrome. [1]

- Inject with antibodies regularly

[Total: 12]

QUESTION 10

Reproduction in seahorses, *Hippocampus*, is unusual as it is the male rather than the female that becomes pregnant. The male has a brood pouch located on its tail. The larger the male the larger the pouch. The female transfers unfertilised eggs into the pouch. The larger the female the more eggs are produced that can be transferred to the brood pouch. The male releases sperm onto the eggs and they are fertilised. The male carries the developing brood for a period of several weeks until he finally gives birth.

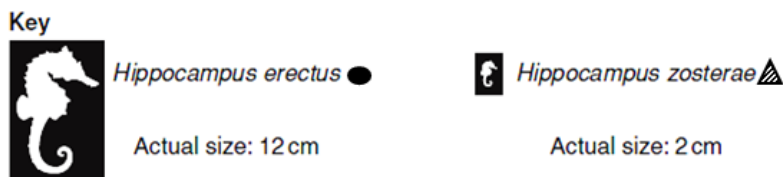
Research into seahorse population has revealed the following:

- They are monogamous. A male and female remain together for the whole mating season.
- Within a population, mates are selected by size. Large females mate with large males and small females mate with small males.
- Few intermediate sized individuals are produced and they have a low survival rate.

Two different species of seahorse are found in the coastal regions shown in Fig. 10.1. The ranges of these two seahorse species overlap in many areas of these waters.



Fig. 10.1



The two seahorse silhouettes are not drawn to scale.

(a) With reference to Fig. 10.1, explain how speciation has occurred in the seahorses. [HI-2] [5]

- There existed variation among the ancestral population of seahorses
- [compulsory] Seahorses undergo sympatric speciation
- **large seahorses mate with large seahorses** and the **small females mate with small males**, hence, they are at **selective advantage**.
- **Intermediate sized seahorses** are at **selective disadvantage** and **do not survive to reproduce** to form fertile and viable offspring.
- Intermediate sized seahorses **do not pass on alleles to the offspring/ only the large and small seahorses can pass on advantageous alleles to offspring**
- **Over many generations, changes in allele frequency of the sizes** in each population occurred
- **Accumulation** of sufficient **genetic differences** over many generations
- [compulsory] Eventually, the different populations were **unable to interbreed** in nature to produce **fertile and viable offspring**, resulting in the formation of different species.
- A: Disruptive selection

(b) There are 4 different species of seahorses from the Atlantic Ocean. Samples of these species were collected and mitochondrial DNA was used to study the evolutionary relationship between the four species.

Explain why mitochondrial DNA is suitable for constructing phylogenetic relationships. [3]

- mtDNA is **present in all eukaryotes**.
- mtDNA has a **high level of genetic variation** in the **non-coding sequences** which can be used to determine phylogenetic relationships among **recently diverged species**.
- The genes of mitochondrial DNA are **well-conserved/ little/slow modifications in the gene**, which enables the determination of the evolutionary relatedness among species that **diverged hundreds of millions of years ago**.
- mtDNA **lacks germline recombination hence** variation in the mtDNA sequence is largely by mutation.
- mtDNA is **inherited through the maternal line**. Effective in tracking only the maternal lineage.
- A larger amount of mtDNA can be extracted from each cell than the nuclear DNA.

Seahorses and spider monkeys have evolved independently to possess a prehensile tail, allowing them to grasp and hold onto structures. Fig. 10.2 shows the prehensile tail structures in both seahorses and spider monkeys.



Fig. 10.2

(c) A student concluded that the prehensile tail shared by both seahorses and spider monkeys is an example of homologous structures. Explain whether the conclusion made is valid. [3]

- **No. It is not valid** [no mark without explanation]
- Both seahorses and spider monkeys develop prehensile tail independently **as they are not evolved from a recent common ancestor**
- OR
- **[Idea of]** Prehensile tail has evolved from different anatomical structures
- It is an example of **analogous structure ...**
- ...which evolved **due to convergent evolution / similar selection pressures in the environment .**

[Total: 11]

• END OF PAPER •